

Development of Infrared Spectroscopy Simulations and Experimental Data Collection for Bis-PEG Materials

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Abstract

This summer, I worked with my faculty mentors on experimental infrared (IR) spectroscopy and computational optical modeling. Experimental efforts focused on expanding the work of previous students by measuring IR spectra for OEG films of chain lengths $N = 11, 12, 13$. Computational work involved developing an optics-based program to simulate IR spectra from theoretical optical constants. This work provided the foundation for a generalizable simulation tool applicable to both OEG films and other thin-film systems.

1 Goals

(1) Extend experimental IR spectroscopy data for OEG films to longer chain lengths. (2) Construct a computational model in Python capable of predicting IR spectroscopy data using reflectance (R) and transmittance (T) for multilayer optical systems.

2 Experimental Component

We collected IR spectra for OEG chain lengths $N = 11, 12, 13$. These compounds were synthesized, and data were collected using procedures developed by previous students. These measurements extend the dataset and enable improved modeling of how molecular length affects spectral features, such as the oscillatory behavior in the intensities of the OEG vibrational peaks. The precursors for longer chain lengths were not available for purchase, so we were unable to synthesize films with chain lengths beyond $N = 13$. See bottom 3 spectra in Figure 1 on page 4.

3 Computational Modeling

3.1 Single-Interface Optics

We began developing code with a function for *Snell's law*:

$$n_i \sin \theta_i = n_t \sin \theta_t. \tag{1}$$

This relates the angle of incidence (θ_i) to the angle of transmission (θ_t) across an interface between media with refractive indices n_i and n_t .

Using the θ_t obtained from *Snell's law*, we created functions for the *Fresnel coefficients*. The Fresnel coefficients describe the amplitude of the electric field that is reflected or transmitted for *s*- and *p*-polarized light at a single interface:

$$r_s = \frac{n_i \cos \theta_i - n_t \cos \theta_t}{n_i \cos \theta_i + n_t \cos \theta_t}, \quad t_s = \frac{2n_i \cos \theta_i}{n_i \cos \theta_i + n_t \cos \theta_t}, \quad (2)$$

$$r_p = \frac{n_t \cos \theta_i - n_i \cos \theta_t}{n_t \cos \theta_i + n_i \cos \theta_t}, \quad t_p = \frac{2n_i \cos \theta_i}{n_t \cos \theta_i + n_i \cos \theta_t}. \quad (3)$$

Using the *Fresnel coefficients*, we created functions for the reflectance and transmittance:

$$R = |r|^2, \quad T = \frac{n_t \cos \theta_t}{n_i \cos \theta_i} |t|^2. \quad (4)$$

We then created an interactive graphing function to display this data.

3.2 Complex Refractive Index and Absorption

To model metallic surfaces as well as absorption, we expanded our definition of the refractive index to be a complex number:

$$\tilde{n} = n + ik, \quad (5)$$

where n is the real component of the refractive index, $i = \sqrt{-1}$, and k is the absorption coefficient. We updated our previous code accordingly.

3.3 Phase Shifts and Evanescent Waves

Redefining the refractive index introduced new properties to our code. The angle θ_t gained a thickness-dependent phase shift and could become imaginary beyond the critical angle. When θ_i exceeds the critical angle, θ_t becomes imaginary and the transmitted wave is evanescent, propagating along the surface of the film. Evanescent waves are central to attenuated total reflection (ATR) spectroscopy. We created plotting functions to display both the phase shift and the decay of the evanescent wave.

3.4 Two-Interface, Three-Layer Systems

Next, we developed a system capable of handling two interfaces. To do this, we calculated the total Fresnel coefficients for the net system:

$$r_{\text{total}} = r_{01} + \frac{t_{01} r_{12} t_{10} e^{2ik_1 d \cos \theta_1}}{1 - r_{01} r_{12} e^{i2k_1 d \cos \theta_1}}$$

where r_{01} , r_{12} , t_{10} , and t_{01} are Fresnel coefficients at the first and second interfaces, and $e^{2ik_1 d \cos \theta_1}$ is the thickness-dependent phase shift in the middle layer.

3.5 Multi-Layer Systems: Transfer Matrix Method

We then generalized the code to handle any number of interfaces using the transfer matrix method. The matrix formalism used a series of transmission matrices D and propagation matrices P to create a transfer matrix M . We could then solve for the total Fresnel coefficients for any number of layers:

$$r_{\text{total}} = \frac{M_{21}}{M_{11}}, \quad t_{\text{total}} = \frac{1}{M_{11}}. \quad (6)$$

3.6 Spectral Dependence

To generate our first spectra, we used data arrays of n , k , and ν values of SiO_2 from a previous study. We instructed our Fresnel coefficient functions to loop over each set of variables and then used the conversion

$$A = \log_{10} \left(\frac{1}{1 - R} \right)$$

to plot the absorbance and generate our first reflection spectra. See Figures 2 and 3 on page 5.

4 Results

The final code:

- Simulates IR spectra for arbitrary multi-layer stacks.
- Accepts wavelength-dependent n and k values.
- Computes R , T , phase shifts, and evanescent field profiles.
- Visualizes results in multiple formats for analysis.

Experimentally, we recorded spectra of OEG films with chain lengths $N = 11, 12, 13$.

5 Conclusion

Our work this summer resulted in a multipurpose IR spectroscopy simulation tool and new OEG spectral data. We plan to build upon the code this school year by adding the ability to handle oriented films and extract n values from k values via the Kramers–Kronig transformation. We hope to publish this code in a peer-reviewed journal.

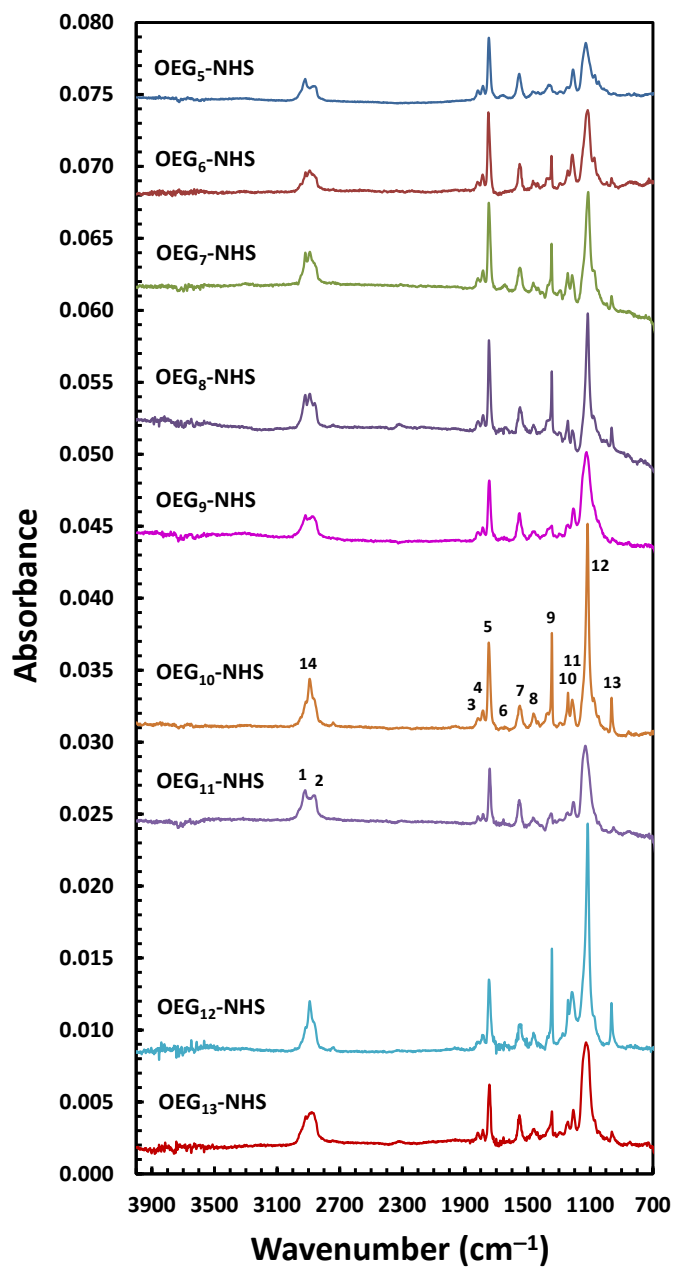


Figure 1: Self-assembled monolayers constructed from 11-mercapto-undecylamine coupled to various chain-length oligo(ethylene glycol)s (OEGs) having N-hydroxysuccinimide (NHS) as the terminal group.

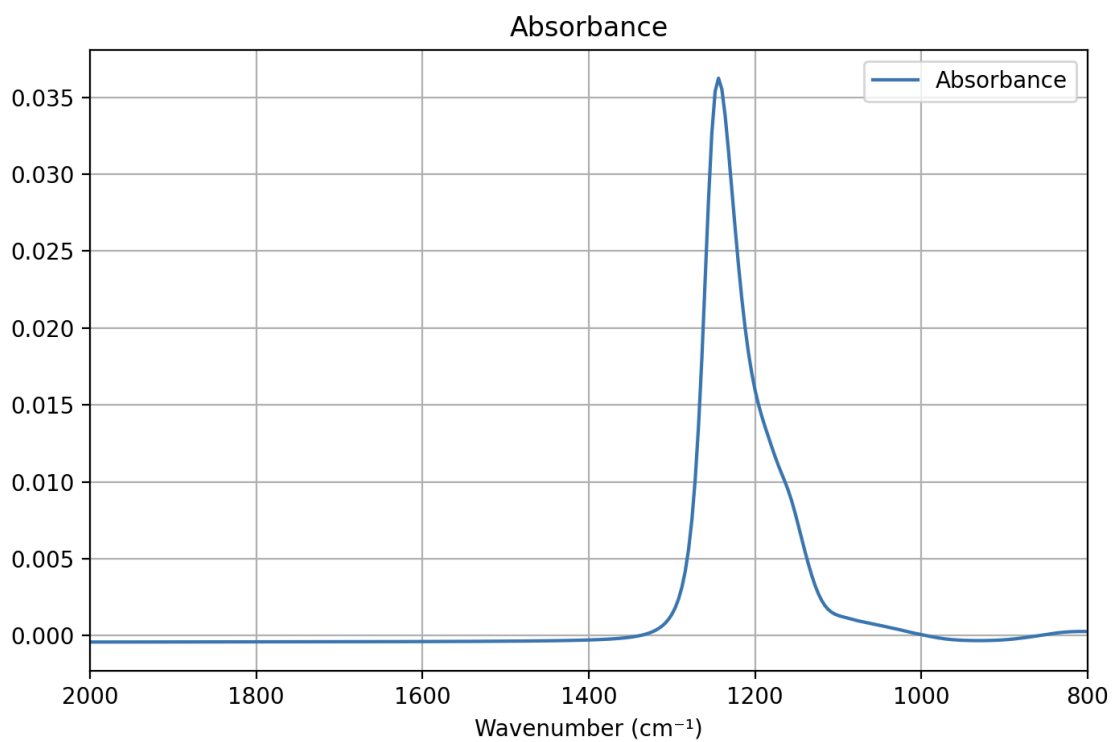


Figure 2: Simulated spectra.

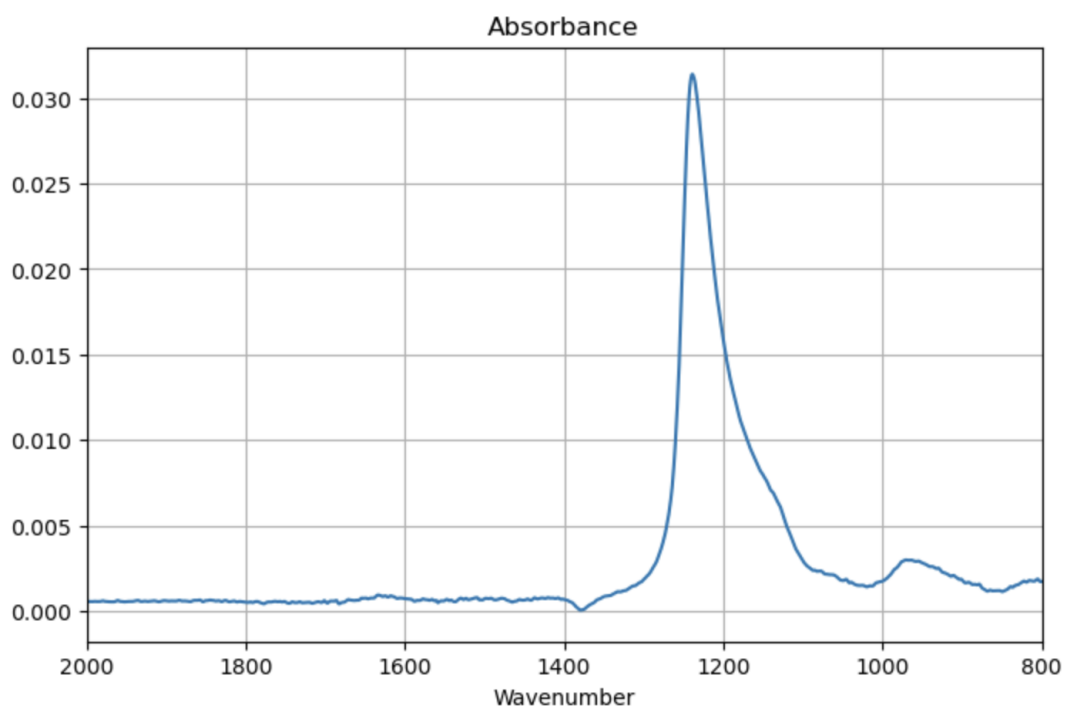


Figure 3: Experimentally obtained spectra.